

AUTONOMOUS DRONE DETECTION & RESPONSE SYSTEM

Final Year Project

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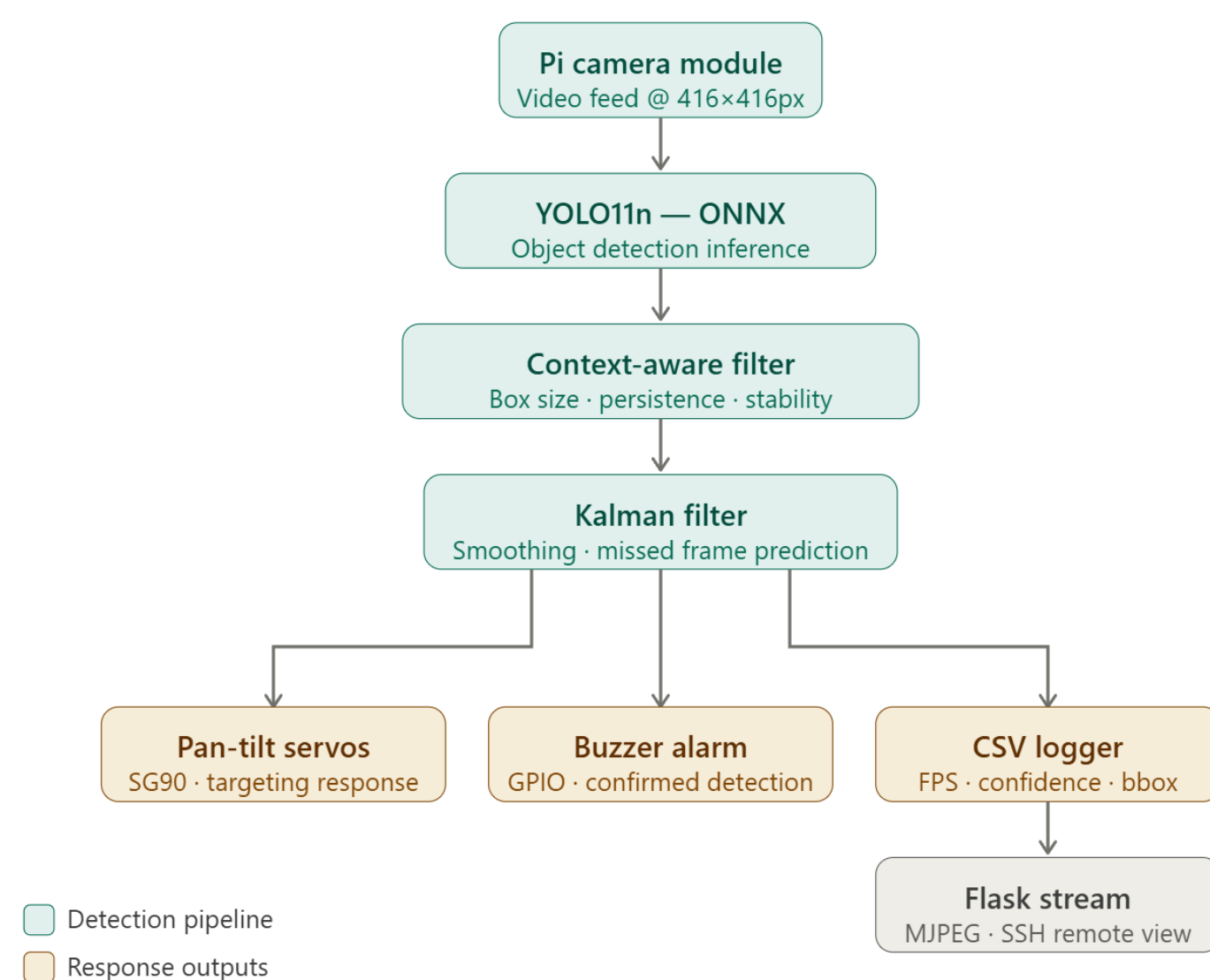
Atlantic Technological University

Bachelor of Engineering (Honours) in Software and Electronic Engineering

Introduction

The motivation behind this work is to address the need for rapidly deployable, temporary surveillance systems operating in dynamic environments where infrastructure is limited. Existing counter-drone solutions demand significant cost and fixed infrastructure, making them impractical for resource-constrained scenarios. This project delivers a self-contained, low-cost drone detection and response system built on a Raspberry Pi 4 – no network required, operational within minutes.

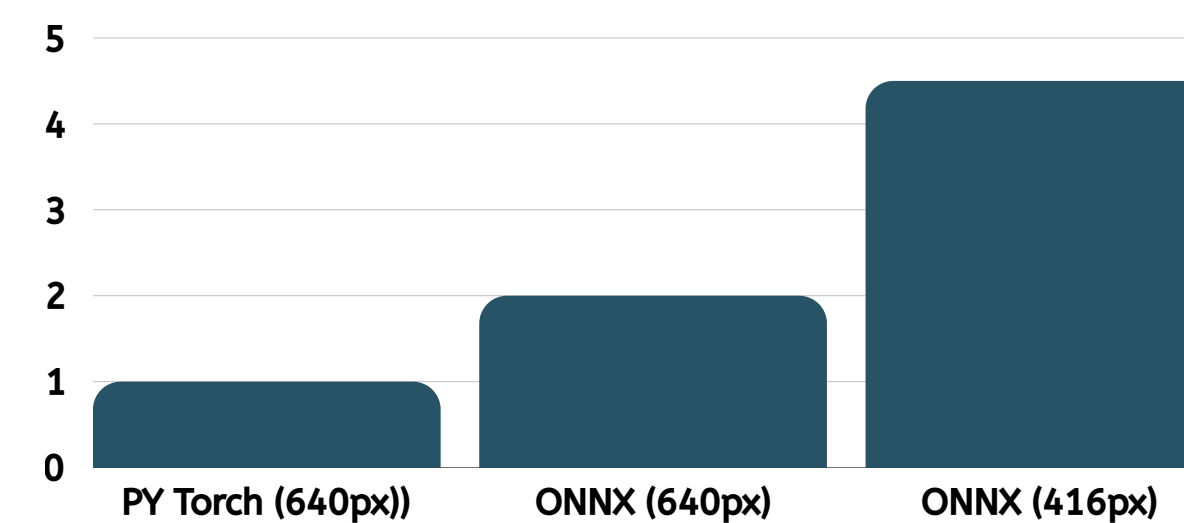
Project Architecture



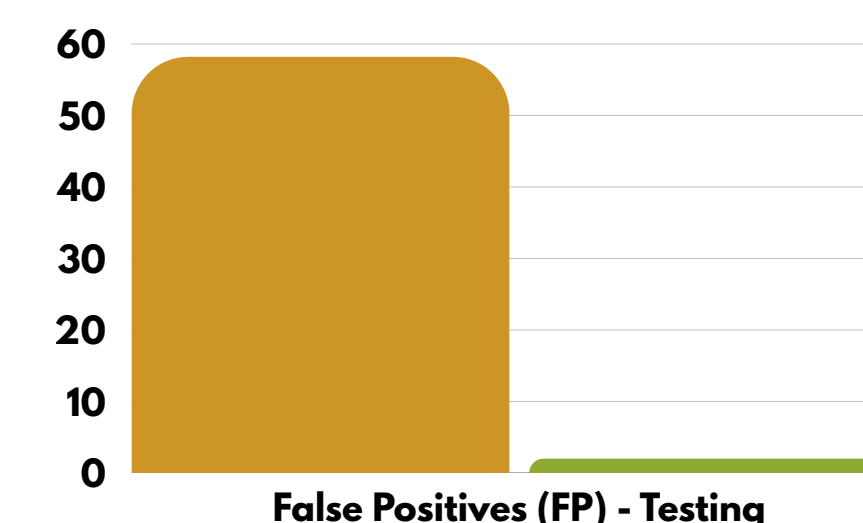
Key Features

- Rapid deployment – self-contained, operational within minutes
- Real-time drone detection using YOLO11n (416x416px inference)
- Deployed on Raspberry Pi 4 – fully embedded, no cloud inference
- Kalman filter for temporal smoothing and missed-frame prediction
- Context-aware filtering: temporal persistence, bounding box size thresholds, box stability, and adaptive confidence thresholding
- Pan-tilt servo system tracks detected drone using bounding box centre coordinates
- GPIO buzzer alarm triggers on confirmed detection

● Frame Rate Per Second (FPS)



● Indoor ● Outdoor



Methodology

1

Model Development - Two datasets were evaluated: a Roboflow drone dataset (baseline) and a DatasetNinja UAV dataset (final deployment model). Both were trained using YOLO11n in Google Colab. The final model was trained for 80 epochs at 416x416px resolution using the AdamW optimiser

2

Embedded Deployment - Model exported to ONNX format for optimised CPU inference on Raspberry Pi 4, operating headlessly via SSH with a Flask MJPEG live stream.

3

Context-Aware Filtering - Raw detections pass through bounding box size thresholds, 3-frame temporal persistence, box stability checking, and adaptive confidence thresholding before a detection is confirmed.

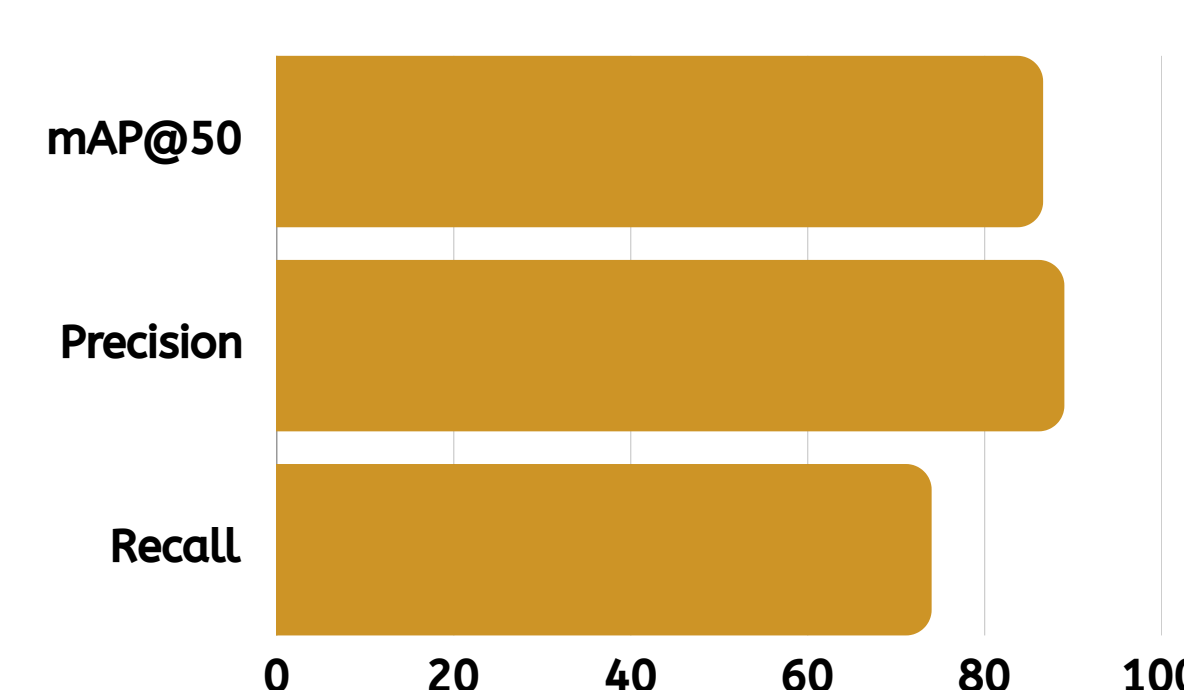
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Kalman Filter Tracking - A linear Kalman filter smooths bounding box positions and maintains tracking continuity across missed detection frames – critical at ~4 FPS.

5

Pan-Tilt Servo Response - On confirmed detection, the Kalman-predicted centre point is mapped to pan and tilt servo positions, with smoothing to reduce jitter.

● Model Performance



Pi Camera Live Stream

Log file: /home/pi/yolo/logs/stream_csv_2026-03-20_18-09-33.csv



Results

The ONNX model at 416px achieved a 4x FPS improvement over the baseline PyTorch model, reaching approximately 4 FPS on the Raspberry Pi 4. Context-aware filtering reduced false positives from 58.2 per minute to near-zero in outdoor conditions. The final trained model achieved a mAP@50 of 0.866 with a precision of 0.89, demonstrating reliable detection with low false alarm rates. Outdoor batch testing confirmed consistent drone detection at 5–10 metres with 60–80% confidence.

Software

- Python
- YOLO11n (Ultralytics)
- Google Colab
- ONNX runtime
- Raspberry Pi OS
- OpenCV
- Jira (Project Management)

Hardware

- Raspberry Pi 4
- Raspberry Pi Camera Module v2
- USB Power Supply
- SanDisk Ultra 32GB Memory Card
- 2 x sg90 Servo Motors
- GPIO Buzzer
- Breadboard + Jumper Wires

Conclusion

This project successfully delivered a self-contained, rapidly deployable drone detection and response system on low-cost embedded hardware. Operating at approximately 4 FPS with near-zero false positives outdoors, the system demonstrates a viable alternative to costly fixed infrastructure solutions. Future work could explore higher-resolution inference on more capable hardware, multi-object tracking, and acoustic sensor fusion to further extend detection range and reliability.

